

PulseEDU – Feature Dependency Warnings

Change summary – School Features admin panel

Date: July 2, 2026 | Area: client (App.tsx) – no database or API changes

This is a single, self-contained UI change plus a dependency audit of the nine newest modules. There are no schema migrations, no server route changes, and no new dependencies to install. Existing behavior is unchanged except for new on-screen guidance.

1. What changed and why

PulseEDU features are gated in two tiers: the district SuperUser “Allowed” switch and the school admin “Enabled” switch. A feature is live only when both are on.

Some features depend on others (for example, the School Store needs PBIS Points). The SuperUser licensing editor already blocked incoherent combinations, but the everyday “School Features” panel – where admins actually flip features on and off – showed no dependency guidance at all. An admin could turn a feature on and not realize a prerequisite was off, leaving the feature silently broken.

This change surfaces those dependencies as inline warnings directly in the School Features panel, so no one is surprised by a feature that quietly does not work.

2. Dependency audit – the nine newest modules

Before adding the warnings, each recently-added module was checked against its actual code to confirm there are no hidden couplings. Findings:

- Gradebook — correctly **REQUIRES** Data Imports (it is an import surface).
- Data Chats — correctly **RECOMMENDS** Academics (uses FAST data, but works without it).
- School Grade Calculator — correctly **RECOMMENDS** Academics.
- Pickup, Event Tickets, School Tours, e-Sign, PulseBrainLab, Safety Plans — genuinely standalone.

Verified specifically: PulseBrainLab uses its own groups (not the Behavior Specialist module), and Event Ticketing email is standalone – Parent Portal delivery is one optional channel, not a hard requirement. No feature silently turns another on or off; the only cascade (plan-apply writing the district gates) is intentional.

3. New on-screen behavior

For each feature that is effectively live (both gates on) whose dependency is off, the panel now shows an inline note under that feature:

- **Red, bold** — a **HARD** requirement is missing; the feature will not work until it is enabled.
- **Amber** — a **SOFT** recommendation is missing; the feature works but is degraded.

Example messages

& Requires PBIS Points — turn it on or this feature won’t work.

Works best with Academics enabled.

*The panel **WARNS** but does not block – it is a plain save, not the plan editor (which does block). Warnings appear only when a feature is actually live, so staging a not-yet-allowed combination stays quiet.*

4. The dependency map (as implemented)

Mirrors the server registry (FEATURE_KEYS in lib/featureLicensing.ts) for the features present in this panel:

SchoolStore	requires	PBIS Points	(HARD)
Gradebook	requires	Data Imports	(HARD)
DataChats	recommends	Academics	(SOFT)
SchoolGrade	recommends	Academics	(SOFT)

Academics and Data Imports are district-level and are not rows in this panel, but their allowed/enabled values are available in the settings state, so the effective-state check still works correctly.

5. Files touched, testing & risk

5.1 Files

- artifacts/client/src/App.tsx — dependency map + inline warnings in the School Features panel (+74 lines).
- (internal) developer memory note documenting this as a new dependency surface.

5.2 Testing

- Client typecheck passes.
- Independent code review: approved (logic matches the runtime gate; map accurately mirrors the server registry).

5.3 Risk

Low. UI-only, additive change inside an already-working panel; no data, API, or gating logic was altered. The one maintenance note: the panel's dependency map is a small hardcoded mirror of the server registry and must be kept in step if feature dependencies change in the future.

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